VitaTwin AI

Predict. Simulate. Prevent.

An AI-powered healthcare Digital Twin proof-of-concept for early glycemic-risk awareness in Type 2 Diabetes

Happiest Health Digital Twin Challenge 2026

Team: A Girl :) | SRM Madurai College for Engineering and Technology

THE PROBLEM

Healthcare today reacts after the event has already happened

DATA

A patient's medical record and wearable data exist — but sit unused between visits.

EVENT

A glucose excursion, or other adverse event, occurs unpredicted.

REACTION

Clinical response happens only after the fact — the costliest point to intervene.

Type 2 Diabetes is one of India's most prevalent chronic, lifestyle-linked conditions — and glycemic instability is a leading driver of avoidable complications.

OUR APPROACH

From reacting to a single event, to living alongside a twin

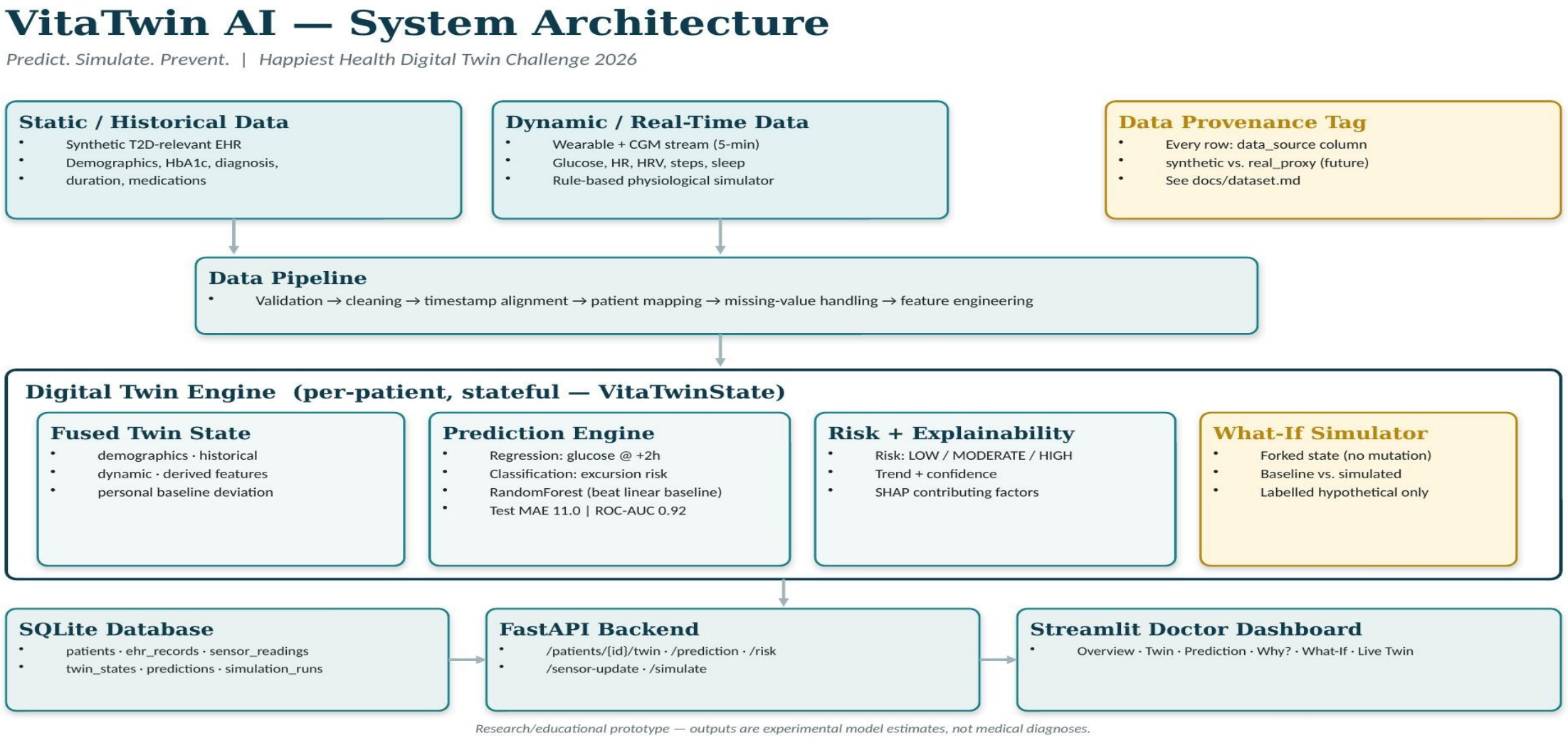


Traditional healthcare: DATA → EVENT → REACTION

VitaTwin AI creates a living computational representation of a patient — combining what we already know with what is happening right now — so the AI can estimate what could happen next, and clinicians can explore hypothetical futures before deciding.

SYSTEM DESIGN

Architecture



DATA STRATEGY

Synthetic, disclosed, and schema-compatible with real data

Static / Historical

Synthetic T2D-relevant patient population
Age 35-72, BMI 23-39, HbA1c 5.8-9.8%
Diagnosis, duration, medication profile

Dynamic / Real-Time

5-minute glucose / HR / HRV / steps / sleep
Rule-based physiological simulator
Meal + activity + circadian response modeling

Why synthetic, and why that's the right call

Real CGM+wearable datasets (OhioT1DM, D1NAMO) are Type 1 diabetes research cohorts — there is no strong open dataset combining T2D-specific EHR with continuous wearable+CGM data. Rather than hide this mismatch, every row carries a data_source tag, and the dynamic-stream schema is built to be drop-in compatible with real datasets via a documented loader stub — so real data replaces the simulator later without touching downstream code.

CORE INNOVATION

An explicit Digital Twin state — not just a model

- VitaTwinState: a real, persisted per-patient object — demographics, historical record, dynamic signals, derived features, prediction, and explanation — not a label a model spits out
- Update loop: new sensor data → validation → timestamp alignment → twin update → feature engineering → prediction → risk → explanation → dashboard refresh
- Personal baseline: risk is judged against each patient's own recent glucose baseline, not only a population average — "current state is 19.7% above this patient's baseline"
- What-If simulation runs on a forked copy of the state — it never mutates the real, persisted twin

MACHINE LEARNING

Baselines first, temporal validation, no leakage

Dual Prediction Target

Regression: glucose value
~2 hours ahead

Classification: probability of a
hyperglycemic excursion
(>180 mg/dL) within 2 hours

Models — Trained & Evaluated

Regression: Ridge vs. Random Forest
-> Random Forest won (val MAE
10.7 vs. 15.9)

Classification: LogReg vs. RF
-> Random Forest won (val ROC-AUC
0.92 vs. 0.85)

Real, run, test-set results -- see
docs/model.md

Validation Discipline

Per-patient chronological
70/15/15 split (train/val/test)

No shuffling -- adjacent 5-min
readings are correlated, so a
random split would leak

Test MAE: 11.0 mg/dL
Test ROC-AUC: 0.92

TRUST & TRANSPARENCY

The clinician never sees a bare risk score

Risk: HIGH

↑ glucose trend (rising 3 of last 4 readings)

↓ recent activity vs. patient baseline

↓ sleep duration last night

↑ 22% deviation from personal glucose baseline

How

SHAP (TreeExplainer) on the selected tree-based model gives signed, per-prediction feature contributions.

We show direction and rank of the top 4-5 factors — never a raw SHAP plot dumped on a clinician, and never a causal claim the evidence doesn't support (e.g. we do not say low sleep "caused" the glucose spike).

KEY DIFFERENTIATOR — REAL OUTPUT, NOT A MOCKUP

What-If Simulator: baseline vs. simulated scenario

Baseline — Patient VT001, actual state

Predicted glucose (+2h): 186 mg/dL
Excursion probability: 81%
Risk category: HIGH

Simulated — +activity, +sleep, -glucose

Predicted glucose (+2h): 179 mg/dL
Excursion probability: 70%
Risk category: HIGH (reduced, not flipped)

Honest finding, stated plainly

Even with a combined activity+sleep+glucose intervention, risk stayed HIGH -- this baseline model is (correctly) dominated by current glucose over short-horizon lifestyle levers. Documented as a limitation and a concrete next step (richer meal/medication features), not hidden. Every number above came from actually running src/simulation/whatif.py.

DEMONSTRATION

Live Twin: a real, working replay engine

- Built and working: a “Live Twin” tab replays each patient's real last 24 hours, 5 minutes at a time, with Play / Pause / Reset / Scrub and adjustable speed (1x / 4x / 12x)
- Each step re-runs the ACTUAL trained model live: Digital Twin updates → prediction recomputes → risk re-assesses → the same risk ring/SHAP panels used everywhere else update in real time
- Verified end-to-end: in one real patient's 24h window, risk genuinely moves LOW → MODERATE → HIGH → MODERATE → LOW as the day progresses — not scripted, just what the model actually outputs at each real historical point
- Jury demo flow: select patient → open Live Twin → press Play → watch risk rise and fall live → pause on a HIGH moment → open “Why this prediction?” → open What-If Simulator

STATUS

Progress & Roadmap

Complete — real, tested code

✓ Phase 0 — Feasibility report
✓ Phase 1 — Synthetic data generator (11/11 tests)
✓ Phase 2 — Preprocessing & fusion pipeline
✓ Phase 3 — Digital Twin engine (fork/update/snapshot)
✓ Phase 4-5 — Trained models, real test metrics
✓ Phase 6 — SHAP explainability
✓ Phase 7 — What-If simulator + recommendations
✓ Phase 8 — FastAPI backend (12 endpoints)
✓ Phase 9 — Two dashboards: Streamlit + custom web UI
✓ Phase 10 — Live Twin replay (real, working, tested)
✓ Phase 11 — Full test suite: 66/66 passing
✓ Phase 12 — Documentation (7 docs, all real)

Remaining

○ Phase 13 — Record the actual demo video
○ (script ready in docs/demo_script.md,
○ every number verified against the
○ live running system)

○ Team name + college name
○ (placeholders in README/deck/LICENSE)

Safety, honestly stated

VitaTwin AI is a research and educational prototype. Predictions are experimental model outputs and are not medical diagnoses or treatment recommendations. All data is synthetic or disclosed as a physiological proxy — no real patient data is used.

Team

A Girl :) — SRM Madurai College for Engineering and Technology
Team Leader: Dharshini M

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